

Interactive Digital Art and Design SE608

Project Proposal Year IV

Department of

Computing

C A R L O W

Student Name:	Kacie Franklin	Student Number:	C00299407
Working Title:	Ax-Iom		
Description:	Ax-Iom is a retro style first person shooter with the player taking the role of an older modified Ax-Iom robot model created by Ion Robotics Inc. The world had fallen into an apocalyptic state due to nanotech, created by the corporation, going haywire. The nanotech in the newer Ax-Iom models began to infect other machinery and eventually spreading to living organisms, animals, plants and humans. The Ax-Iom model the player controls, AX-524A, is an altered prototype of an outdated model, lacking a more human appearance, voice, movements, thoughts and manufactured emotions to be more friendly for humans, programmed with the sole function to destroy any infected entities. The game will open with a cutscene of AX-524A booting online and disconnecting from the wires it was attached where the player will then be given a small tutorial for movement, swapping weapons, interacting with the environment and shooting throughout the tutorial level of the abandoned robotics lab. The game will allow the player to collect different hearts, when a boss is defeated, which will give the player different perks, positive or negative when equipped and give different durability (health). The player will start with a saw as their melee weapon for the entire game and gaining new weapons as they play. There will be three levels before each boss fight, with the game having four bosses total, the boss will give the player its heart when defeated or at least what counts as its heart. In between boss fights the player can interact with survivors in a derelict building to talk and make their own dialogue options that can alter the ending depending on what choice the player makes. I am planning to use a mix of 2D and 3D graphics for the game, I'm considering using 3D for the environment and 2D for sprites, but the bosses will be made in 3D as opposed to being 2D like the default enemies. This isn't concrete yet so that might change and be fully 3D instead. I intend to make my own soundtrack for the game as well, but I haven't decided on the software for it yet, I've used GarageBand to play around and		

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	make my own tracks before but I would be willing to try a few other software to make the game soundtrack with.
Reasons for selecting project:	I chose this as my project because I'd like to improve my skills with 3D modelling, using Maya or Blender to create game assets, add my own textures, understand how animation works in the respective software. I would also like to improve on my 2D art, making concepts for levels, drawing things I have never drawn before and overall improve with my art and animation. I have also never used a 3D game engine before, or have made a 3D game, so I believe it would broaden my skillset by learning its documentation and creating a game using it. I could also add it onto my CV that I know my way around the engine and create and rig 3D assets.
Proposed research content:	I plan to play Doom and Ultrakill, two retro style shooters, as they are what inspired this project, to gain a better understanding of gameplay and mechanics as well as to decide what way I want to do the overall graphics for my own game as part of my research when I have the time. I will be using this study as a guideline for my game as I don't want to make it too difficult for people who don't play shooters and I don't want to make the difficulty too easy for people who are avid enjoyers of the genre. An Analysis of How First-Person Shooting Games Retain Players of All Skill Levels I will also involve this paper on level design in first person shooters as I will need to design each level for my game. Design Patterns in FPS Levels This paper goes over the use of more old-style graphics in modern games which I intend to use in my own project due to liking the older style of graphics over the hyper realistic graphics some games use now, likely due to nostalgia. Retro Stylistic Transformation in Games

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	It's not a paper but I intend to research how shaders work as I will be using them to give my game a more pixilated aesthetic to appear more retro. Shader Development
External links (if applicable):	I found a tutorial online for how to add inputs from a PS4 controller to the player controller which requires a plugin to achieve: PS4 Controller Input Tutorial I also found an entire series of tutorials that shows how to create a retro style shooter in Unreal Engine 5 which I will be using to help throughout creating my own game for various basic mechanics if I need them: How to Make a Retro Style FPS Game in Unreal Engine 5 Including the documentation for Unreal Engine here: Unreal Engine 5 Documentation
Hardware requirements:	D428 computer PS4 controller (owned) iPad and apple pen (owned) Personal Laptop(owned)
Software requirements:	Unreal Engine 5 Maya or Blender Procreate(owned) GarageBand(owned) Substance Painter
Other requirements:	The tutorial for the PS4 controller inputs requires the IP and Vender of the pc to work, so I might need to set that up again and export the game again for it to work with the D428 computer I'll be using to showcase it. I'll check that when I have basic mechanics working to confirm if that is the case or not or try out different tutorials.

Signed: Kacie Franklin

Date: 20/09/2026

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Reasons for not approving project:		
Conditions attached to approving project:		
Approved/Not Approved:		
Name of Supervisor:		
Signed: _____	Date: _____	